Gravity-Constrained Post-Training of JEPA World Models

Method note

We post-train the predictor of a pretrained JEPA world model so that, for any gravity it is conditioned on, the object motion encoded in its predicted latents follows free fall. Physics enters through a single constraint, evaluated with a frozen position probe on rollouts under counterfactual gravities for which no video exists. A switching gradient update alternates between restoring feasibility and decreasing the JEPA loss.

1 Problem setting

Each video shows K objects moving under a uniform gravitational acceleration $g \in \mathbb{R}^2$, constant within an episode and varying across episodes; the training episodes cover a limited range of g. Gravity is the only force acting: every window used below ends before the first contact. We seek a predictor whose rollouts follow free fall under any g it is conditioned on, including values outside the training range, without increasing its prediction error on real videos. The main text assumes that g is given for each training episode; Appendix **??** removes this assumption.

Symbol	Meaning
x_t, Δt	video frame at step t; frame interval
E, $z_t = E(x_t)$	pretrained encoder (frozen); latent of frame t
P_θ, $\hat{z}_{t+s}$	predictor with trainable parameters θ; its s-step prediction from $z_{\le t}$
g, g', $\mathcal{G}$	gravitational acceleration of a real episode; counterfactual gravity; set from which g' is sampled
h, $p_{k,\tau}$, K	position probe (frozen); position of object k decoded from the latent at step τ; number of objects
$a_{k,\tau}$, $\bar{g}_\tau$	finite-difference acceleration of object k at step τ; its target
H	rollout horizon ($H \ge 2$)
$\mathcal{L}$, c, C	expected JEPA loss; violation of one rollout; expected violation
$\mathcal{L}_B$, C_B	minibatch estimates of $\mathcal{L}$ and C
ε, ρ	tolerance on C; its relative margin over real data
$\hat{C}$, β	exponential moving average of C_B, used to select the mode; its decay rate
d, η	update direction; step size

2 Preliminaries

JEPA world model. The encoder maps frames to latents, $z_t = E(x_t)$, and the predictor rolls out autoregressively in latent space, $\hat{z}_{t+1:t+H} = P_\theta(z_{\le t}, g)$ [1]. Its training loss is

$$\mathcal{L}(\theta) = \mathbb{E}\Big[\sum_{s=1}^{H} \big\|\hat{z}_{t+s} - z_{t+s}\big\|^2\Big], \tag{1}$$

with the expectation over real windows and their gravity. The encoder is frozen, so the targets z_{t+s} carry no gradient. We start from a pretrained model; if its predictor takes no gravity input, a FiLM adapter [2] initialised to the identity is added, so that post-training starts exactly from the pretrained predictor.

Free fall in discrete time. The only physics used is the following identity.

Lemma 1. *Let an object have position p_t and velocity v at step t, and move freely under gravity g before t and g' after t, so that for every $s \ge 0$*

$$p_{t+s} = p_t + v\,s\Delta t + \tfrac{1}{2}g'(s\Delta t)^2, \qquad p_{t-s} = p_t - v\,s\Delta t + \tfrac{1}{2}g(s\Delta t)^2.$$

Then the finite-difference acceleration $a_\tau = (p_{\tau+1} - 2p_\tau + p_{\tau-1})/\Delta t^2$ satisfies

$$a_t = \tfrac{1}{2}(g + g'), \qquad a_\tau = g' \text{ for } \tau \ge t+1,$$

independently of v. For $g' = g$ both reduce to $a_\tau = g$.

Proof. For $\tau \geq t+1$ the positions $p_{\tau-1}, p_\tau, p_{\tau+1}$ are all given by the first formula; the constant and linear terms cancel in the second difference and the quadratic term contributes $g'\Delta t^2$. At $\tau = t$, $p_{t+1} - 2p_t + p_{t-1} = (v\Delta t + \frac{1}{2}g'\Delta t^2) + (-v\Delta t + \frac{1}{2}g\Delta t^2)$. □

Position probe. The probe h is a small network trained on top of the frozen encoder to regress the positions of the K objects, using frames with known positions (simulator states or tracker labels), and then frozen [3]. We write $p_{k,\tau}$ for the position of object k decoded by h from the latent at step τ, whether that latent is encoded from a real frame or predicted. Two requirements apply. (i) The held-out error of h must be small compared with $\|g\|\Delta t^2$, the displacement gravity adds in one step; otherwise the finite-difference accelerations are dominated by noise. (ii) A second, independently trained probe, never used during post-training, is reserved for evaluation, so that a predictor that satisfies the constraint only by exploiting h can be detected.

Switching gradient method. For $\min_\theta \mathcal{L}(\theta)$ subject to $C(\theta) \leq \varepsilon$, the switching (sub)gradient method [4] steps along $-\nabla C$ while the constraint is violated and along $-\nabla\mathcal{L}$ otherwise. Stochastic variants [5] and extensions to weakly convex problems [6] come with complexity guarantees. These guarantees do not extend to deep networks, where the method serves as a principled heuristic.

3 Method

Counterfactual rollouts. Given a real window with latents $z_{\leq t}$ and gravity g, sample a counterfactual gravity $g' \sim \mathcal{U}(\mathcal{G})$, where $\mathcal{G}$ is wider than the training range, and roll the predictor out under it: $\hat{z}_{t+1:t+H} = P_\theta(z_{\leq t}, g')$. No video of this future exists, but Lemma 1 specifies how its positions must evolve. Because the state at t is real, the predictor and the probe receive in-distribution inputs at the start of the rollout; only the conditioning gravity changes.

Definition 1 (Free-fall violation)**.** Decode $p_{t-1} = h(z_{t-1})$ and $p_t = h(z_t)$ from the real latents and $p_{t+s} = h(\hat{z}_{t+s})$, $s = 1, \ldots, H$, from the predicted ones, and let $a_{k,\tau} = (p_{k,\tau+1} - 2p_{k,\tau} + p_{k,\tau-1})/\Delta t^2$. With the targets $\bar{g}_t = \frac{1}{2}(g+g')$ and $\bar{g}_\tau = g'$ for $\tau > t$,

$$c = \frac{1}{KH}\sum_{k=1}^{K}\sum_{\tau=t}^{t+H-1} \left\| a_{k,\tau} - \bar{g}_\tau \right\|, \qquad C(\theta) = \mathbb{E}[c], \tag{2}$$

where the expectation is over real windows and $g' \sim \mathcal{U}(\mathcal{G})$.

The term at $\tau = t$ is essential. Position and velocity are continuous when the gravity changes, so the first predicted step must continue the observed motion; by Lemma 1 this is equivalent to $a_{k,t} = \frac{1}{2}(g+g')$. Without this term, a rollout with the correct curvature but an incorrect initial velocity would incur zero violation.

Proposition 1. *If $c = 0$, the decoded positions coincide with the exact counterfactual trajectory: $p_{k,t+s} = p_{k,t} + v_k\, s\Delta t + \frac{1}{2}g'(s\Delta t)^2$ for $s = 1, \ldots, H$, where $v_k = (p_{k,t} - p_{k,t-1})/\Delta t + \frac{1}{2}g\Delta t$ is the velocity at step t implied by Lemma 1.*

Proof. If $c = 0$, every second difference is fixed, so the recursion

$$p_{k,\tau+1} = 2p_{k,\tau} - p_{k,\tau-1} + \Delta t^2\, \bar{g}_\tau, \qquad \tau = t, \ldots, t+H-1,$$

determines $p_{k,t+1}, \ldots, p_{k,t+H}$ uniquely from $p_{k,t-1}$ and $p_{k,t}$. By Lemma 1 the exact counterfactual trajectory satisfies the same recursion with the same initial values. □

The constraint therefore encodes the complete free-fall law in probe coordinates, not merely a necessary condition. It acts on positions alone; all other content of the latent is governed by the JEPA loss.

Tolerance. Probe noise makes $c > 0$ even for real motion. Let c_{real} denote c evaluated on held-out real windows with the real future latents $z_{t+1:t+H}$ in place of predictions and $g' = g$. We set

$$\varepsilon = (1+\rho)\,\mathbb{E}[c_{\text{real}}], \qquad \rho \approx 0.25. \tag{3}$$

[Qijia: used to exclude the measurement error introduced by the position probe.]

The true dynamics thus satisfy the constraint in expectation, so enforcing it does not bias the predictor away from them, provided the probe is as accurate on counterfactual states as on real ones. This requires counterfactual rollouts to stay within the range of positions and velocities covered by the probe's training data, which is one reason to keep H short.

Why counterfactual gravities are needed. The JEPA loss (1) requires future frames, so it informs the predictor only about the gravities present in the data. On those gravities the constraint adds little: real motion already follows free fall, and predicting the real future already implies the correct accelerations. The constraint requires no future frames and can be evaluated for any $g' \in \mathcal{G}$; it is the only training signal the predictor receives about gravities outside the training range.

Switching update. The expectations $\mathcal{L}$ and C cannot be evaluated exactly. Each step draws a minibatch B of real windows, each with its own g', and uses the minibatch means $\mathcal{L}_B$ and C_B; their gradients are unbiased estimates of $\nabla_\theta \mathcal{L}$ and $\nabla_\theta C$, so the update is a stochastic switching gradient method. A single C_B is too noisy to select the mode, so the mode is selected by $\hat{C}$, an exponential moving average of C_B with decay rate $\beta = 0.9$. In short, C is the constrained quantity, C_B supplies the gradient, and $\hat{C}$ selects the mode.

The mode switches to RESTORE when $\hat{C} > \varepsilon$ and back to OPTIMISE only once $\hat{C} < 0.8\,\varepsilon$; this hysteresis prevents switching at every step. In RESTORE, $d = \nabla_\theta C_B / \varepsilon$. In OPTIMISE, $d = \nabla_\theta \mathcal{L}_B$, unless the constraint is close to active ($\hat{C} > 0.5\,\varepsilon$) and the two gradients conflict ($\langle \nabla_\theta C_B, \nabla_\theta \mathcal{L}_B \rangle < 0$); in that case the conflicting component is removed by a one-sided PCGrad projection [7]:

$$d = \nabla_\theta \mathcal{L}_B - \frac{\langle \nabla_\theta C_B, \nabla_\theta \mathcal{L}_B \rangle}{\|\nabla_\theta C_B\|^2} \nabla_\theta C_B. \tag{4}$$

The parameters are updated by $\theta \leftarrow \theta - \eta\, d$. The projection (4) is invariant to rescaling $\nabla_\theta C_B$, so the normalisation by ε matters only in RESTORE. If the norms of $\nabla_\theta C_B / \varepsilon$ and $\nabla_\theta \mathcal{L}_B$ still differ by orders of magnitude, RESTORE can be given its own step size.

Proposition 2. *With d from* (4)*, $\langle \nabla_\theta C_B, d \rangle = 0$ and $\langle \nabla_\theta \mathcal{L}_B, d \rangle = \|\nabla_\theta \mathcal{L}_B\|^2 - \langle \nabla_\theta C_B, \nabla_\theta \mathcal{L}_B \rangle^2 / \|\nabla_\theta C_B\|^2 \geq 0$. Hence, to first order in η, the step leaves C_B unchanged and does not increase $\mathcal{L}_B$ on the current minibatch.*

Proof. Both equalities follow by taking inner products with (4); the inequality is Cauchy–Schwarz. The first-order changes of C_B and $\mathcal{L}_B$ are $-\eta \langle \nabla_\theta C_B, d \rangle$ and $-\eta \langle \nabla_\theta \mathcal{L}_B, d \rangle$. □

The guarantee holds only to first order and on the current minibatch; with a finite step and on other data, C can still increase, which the RESTORE mode then corrects.

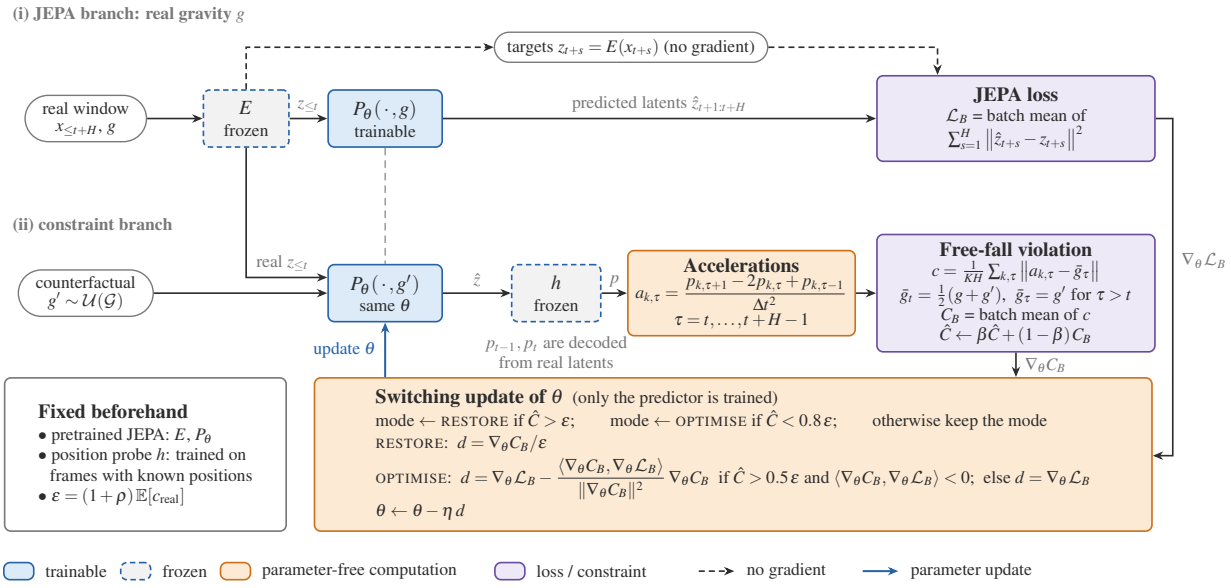


Figure 1: Post-training. Branch (i) is the JEPA loss on real windows conditioned on their real gravity g. Branch (ii) starts from the same real latents, rolls the predictor out under a counterfactual gravity g', decodes object positions with the frozen probe h and measures how far their finite-difference accelerations deviate from the free-fall targets $\bar{g}_\tau$. The switching update combines the two gradients; only θ is trained.

[Qijia: We consider the simplest case where we only require to estimate gravity.]

[Qijia: (i) High-level idea: minimize the JEPA loss subject to physical constraints. (ii) Method: iteratively minimize the JEPA loss, switching to reducing physical constraint violations when they exceed a tolerance. (iii) To detect violations of the physical constraints, we train a position probe to extract position information from the latent space. (iv) The physical law is enforced by constraining the accelerations computed from predicted positions to match the free-fall targets under counterfactual gravity.]

Algorithm 1 Gravity-constrained post-training

Require: pretrained E (frozen) and P_θ; frozen probe h; real windows with gravity labels; set $\mathcal{G}$; tolerance ε; horizon $H \ge 2$; step size η; $\beta = 0.9$

$\hat{C} \leftarrow$ mean of C_B over a few minibatches; mode $\leftarrow$ RESTORE if $\hat{C} > \varepsilon$, else OPTIMISE
repeat
  sample a minibatch B of real windows $(x_{\le t+H}, g)$; $z \leftarrow E(x)$
  $\mathcal{L}_B \leftarrow$ mean over B of $\sum_{s=1}^{H} \|\hat{z}_{t+s} - z_{t+s}\|^2$, with $\hat{z}_{t+1:t+H} = P_\theta(z_{\le t}, g)$
  sample $g' \sim \mathcal{U}(\mathcal{G})$ for each window; $\hat{z}_{t+1:t+H} \leftarrow P_\theta(z_{\le t}, g')$
  $p_{t-1}, p_t \leftarrow h(z_{t-1}), h(z_t)$; $p_{t+s} \leftarrow h(\hat{z}_{t+s})$ for $s = 1, \ldots, H$
  $C_B \leftarrow$ mean over B of c in (2); $\hat{C} \leftarrow \beta \hat{C} + (1-\beta) C_B$
  if $\hat{C} > \varepsilon$ **then** mode $\leftarrow$ RESTORE **else if** $\hat{C} < 0.8\,\varepsilon$ **then** mode $\leftarrow$ OPTIMISE
  if mode $=$ RESTORE **then**
    $d \leftarrow \nabla_\theta C_B / \varepsilon$
  else
    $d \leftarrow \nabla_\theta \mathcal{L}_B$
    if $\hat{C} > 0.5\,\varepsilon$ **and** $\langle \nabla_\theta C_B, \nabla_\theta \mathcal{L}_B \rangle < 0$ **then**
      $d \leftarrow d - \dfrac{\langle \nabla_\theta C_B, d \rangle}{\|\nabla_\theta C_B\|^2} \nabla_\theta C_B$ // *Eq.* (4)
  $\theta \leftarrow \theta - \eta\, d$
until the training budget is exhausted

[Qijia: For the unknown gravity environment, we can estimate the gravity using history windows]

[Qijia: For cases where external forces intervene, we only need to change the gravity to parameters such as a with $F = ma$.]

Gradients reach θ along $c \to a \to h \to \hat{z} \to P_\theta$ and $\mathcal{L}_B \to \hat{z} \to P_\theta$; E and h are never updated. OPTIMISE requires two backward passes (for $\mathcal{L}_B$ and C_B); when $\hat{C} \leq 0.5\,\varepsilon$ the second can be skipped.

4 Evaluation

- Acceleration error of rollouts as a function of g, inside and outside the training range, measured with the held-out evaluation probe, so that exploiting h does not count as success.
- Latent prediction error on real windows, to verify that post-training does not degrade the pretrained model.
- Error of $\hat{g}$ as a function of the context length.
- Ablations: (a) the pretrained model; (b) continued JEPA training for the same number of steps; (c) the constraint with g' restricted to the real gravity of each window (no counterfactuals); (d) the full method. If (c) matches (d), the gain does not come from counterfactual gravities.

References

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